



RB003 Week 5 — English Worksheet

Topic: Buy Buttons — Spend Cash to Unlock Upgrades · **Course:** Tycoon Engineer · **Time:** about 45 minutes

This week your tycoon gets **buy buttons** — touch pads that take the player's cash and unlock a new upgrade by **cloning** a model. A **debounce** makes sure one step on the pad means one buy.

1 · Predict 🧠

Read each script. Before you imagine it running in Studio, write what you think will happen.

```
local price = 50

if cash.Value >= price then
    cash.Value -= price
    print("Bought!")
end
```

The player has exactly 50 cash. Does "Bought!" print? How much cash is left after?


```
local ServerStorage = game:GetService("ServerStorage")
local upgradeModel = ServerStorage.SuperDropper

local model = upgradeModel:Clone()
model.Parent = workspace
```

What appears in the game world? Does the original SuperDropper leave ServerStorage?


```
local buying = false

pad.Touched:Connect(function(hit)
    if buying then return end
    buying = true

    print("One buy!")

    task.wait(1)
    buying = false
end)
```

A player stands on the pad, and **Touched** fires 10 times in one second. How many times does "One buy!" print? Why?

2 · Spot the Bug 🐛

Each script below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The buy button should work when the player has **exactly** the price. A player with exactly 100 cash should be able to buy.

```
local price = 100

if cash.Value > price then
    cash.Value -= price
    local model = upgradeModel:Clone()
    model.Parent = workspace
end
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The buy button should **take the player's cash** when they buy. Right now the upgrade is free — players can buy forever and never lose cash.

```
local price = 100

if cash.Value >= price then
    local model = upgradeModel:Clone()
    model.Parent = workspace
end
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — One step on the pad should mean **one** buy. Right now one step can buy two or three upgrades and take the cash every time.

```
pad.Touched:Connect(function(hit)
  if cash.Value >= price then
    cash.Value -= price
    local model = upgradeModel:Clone()
    model.Parent = workspace
  end
end)
```

Hint: add a **debounce** — a **buying** variable that blocks the pad while a buy is happening.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your tycoon in Roblox Studio. Build a buy button that takes cash and unlocks an upgrade by cloning it. When you finish, come back here.

Send a photo or video of your buy button working, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I ...

My buy button costs ... and unlocks ...

I used **Clone** when ...

One tricky moment was when ...

To fix it, I ...

If I had more time, I would ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while you test your buy button. Talk through it like you are teaching someone who has never seen it. Try to use these words: **buy button**, **price**, **clone**, **debounce**, **subtract**.

1. Show your cash, then step on the buy button.
2. Show the cash going down and the new upgrade appearing.
3. Read your `if cash.Value >= price` line out loud and say what it checks.
4. Say in your own words what a **debounce** does.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.